Forecasting Beijing's PM2.5 six hours ahead

Gradient boosting with leak-free backtesting on the UCI Beijing PM2.5 data, 2010 to 2014

Why it matters: Beijing swings between clean air and severe multi-day episodes, and a forecast a few hours ahead supports health warnings

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Five years of hourly readings, with gaps only in the target

UCI Beijing PM2.5: US Embassy PM2.5, Beijing Capital airport weather

Variable	Unit	Note
PM2.5	µg/m³	target
Dew point	°C	gives humidity
Temperature	°C	
Pressure	hPa	
Wind direction	category	NE, NW, SE, cv (calm or variable)
Cumulative wind speed	m/s	running total, decoded
Snow, rain hours	hours	counters, made flags

Data quality, all years

43,824 hourly rows, none skipped

Only PM2.5 missing: **2,067 h, 4.72%**

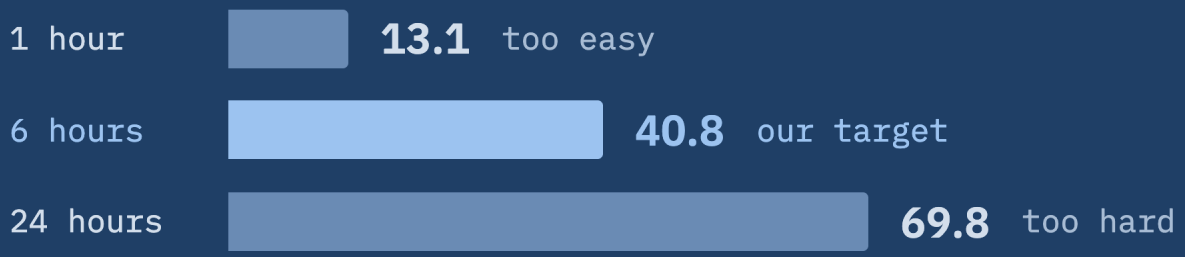
214 gaps; 20 over a day hold 65%

Fill cost, 2010 to 2013: 13 µg/m³ at 1 h, 27 at 3 h

Real extremes: 994 at Lunar New Year

We predict PM2.5 six hours ahead, and must beat the guess “no change”

If we just guess “no change”, how far off are we? Average miss in µg/m³, 2010 to 2013



Six hours is the useful middle. We fixed it from the data before building any model.

The model gets

PM2.5 up to now, weather up to one hour ago

It predicts

PM2.5 six hours from now

The guess to beat (persistence)

“No change”: PM2.5 stays as it is now

Scored by

Average miss (MAE), big misses (RMSE), R^2

Smog builds up over days and often clears within hours once north-west wind sets in



December 2010; shaded hours have north-west wind

Episode: 6+ hours in a row above 150 µg/m³, the “heavily polluted” level. We found 322 in 2010 to 2013.

We use episodes in the physics inputs and the MoE model

Night peak in winter, morning peak in summer

So the model gets the hour and month

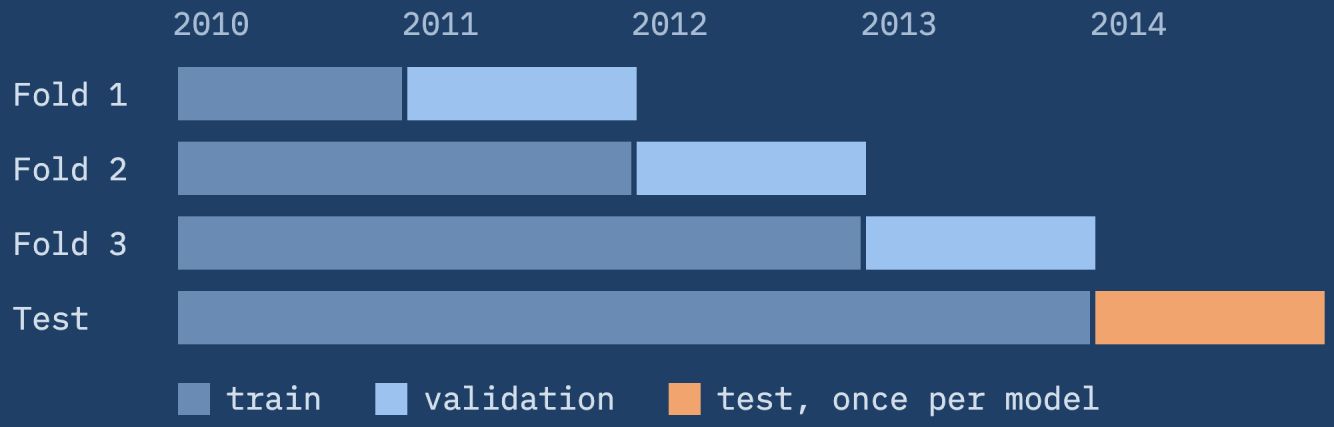
Humid air goes with dirtier air

So the model gets humidity

New Year fireworks: 7 to 14 times the usual level

So the model gets a New Year flag

The model is never shown the year it is tested on



Each fold trains only on earlier years and is validated on the next one. All settings are picked on the validation years, 2011 to 2013.

Missing PM2.5: copy the last value for at most 3 hours

Weather from one hour before, to be safe

A one-day gap between training and test years

Each model takes the 2014 test once; repeats are blocked

Research on Beijing smog gave four new inputs; two rank 3rd and 6th of 52

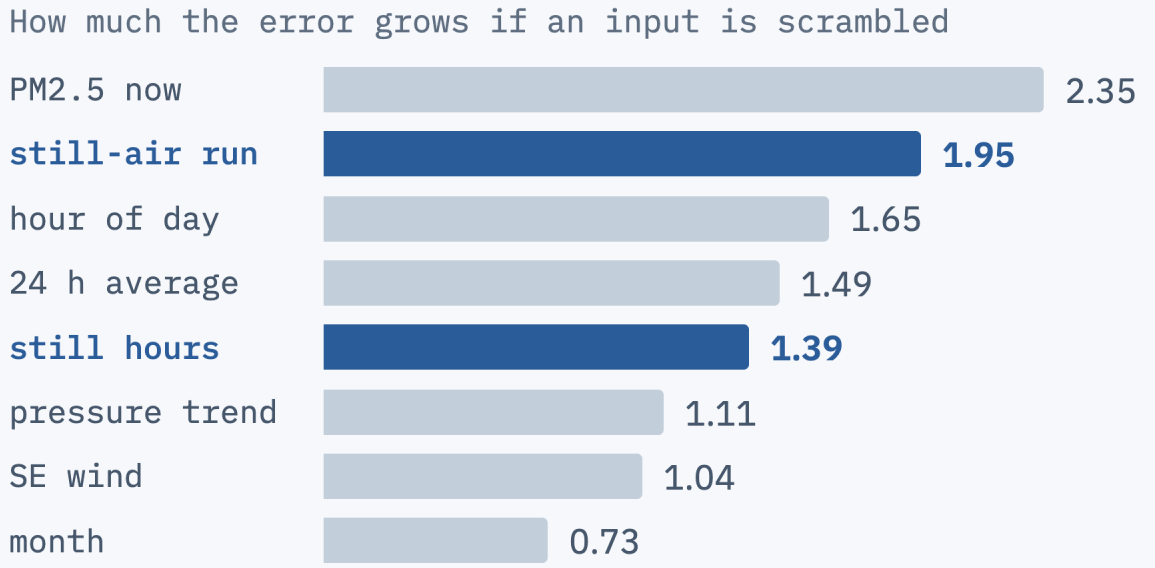
How smog builds (Zhong et al., 2018)

Pollution blows in on southerly winds, then piles up when the air is still and humid

Still-air rule (Huang et al., 2018)

Air counts as still when the wind is below 3.2 m/s and there is no rain

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still hour = wind < 3.2 m/s and no rain
still hours = count in the last 24 h
still-air run = how long it is still
SE wind run = SE wind over last 24 h
humid hours = above 80% humidity, 24 h
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RMSE rise in µg/m³, ranks 2 to 9. 1st is how far PM2.5 now is from its usual level at that hour (19.32). Research inputs in blue. One run, tested on 2013.

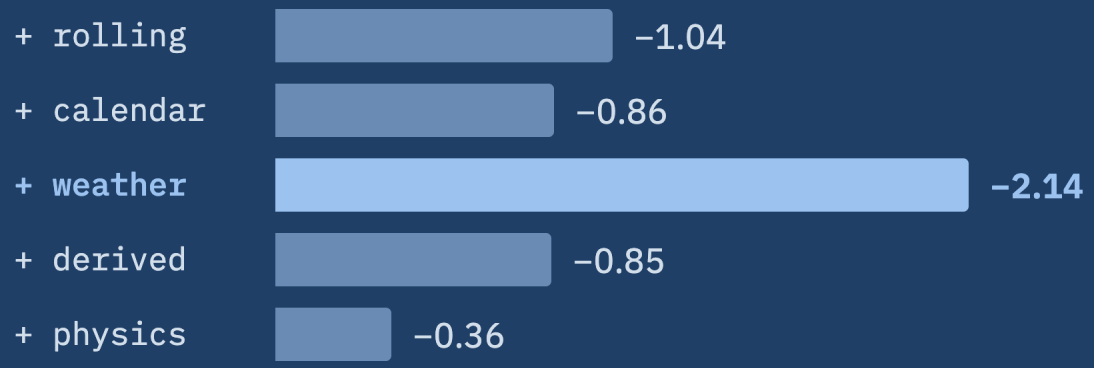
Of the groups added to PM2.5 history, weather lowers the error most

52 inputs, all known at hour t

PM2.5 lags	now and 7 lags, 1 to 48 h	10
Rolling	mean, max, spread, trend	15
Calendar	of the target hour	7
Weather	hour before the forecast	11
Derived	humidity, pressure trends	5
Physics	stagnation, transport	4

RMSE drop as each group is added, µg/m³

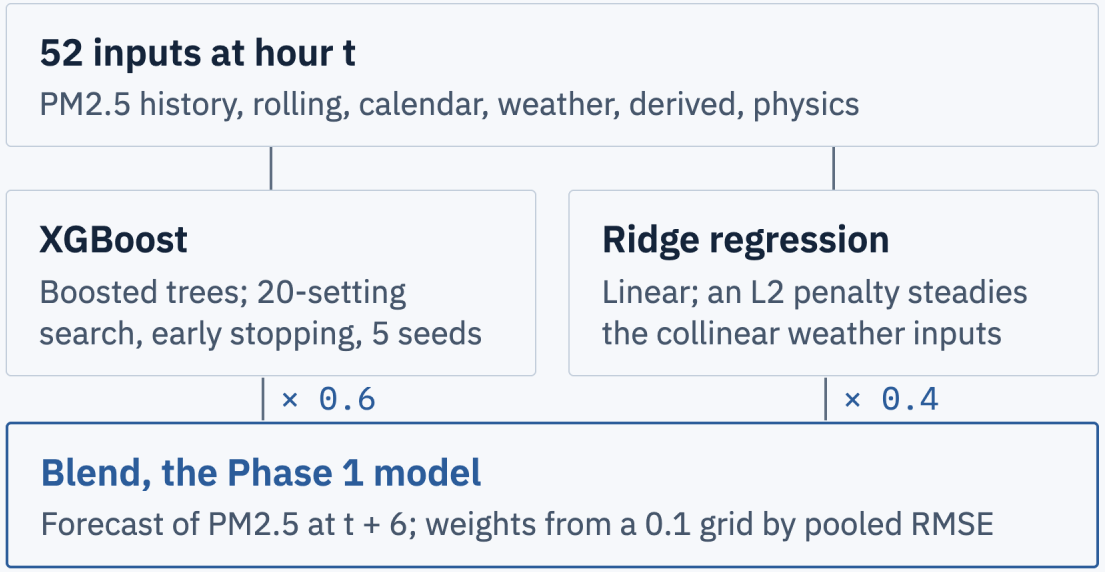
PM2.5 lags alone: **62.05**



All 52 inputs: **56.81**

XGBoost at 6 h, mean of 5 seeds, 2011 to 2013; drops depend on the order added

Ridge and XGBoost each beat persistence; blending them is best



Baselines: PM2.5 now, and the same hour a day earlier

Six hours ahead, 2011 to 2013

Model	RMSE	MAE	R^2
Persistence	65.77	40.23	0.483
Seasonal-naive	101.42	69.42	−0.230
Ridge	57.66	38.27	0.602
XGBoost	56.81	36.46	0.614
Blend	56.33	36.41	0.621

24,417 hours, the same for every model

The sweep confirms six hours; validation picks a raw target

Variance explained (R^2) by horizon, XGBoost one seed

Horizon	Persistence	XGBoost
1 h	0.931	0.939
6 h	0.483	0.614
12 h	0.112	0.365
24 h	−0.229	0.169

The longest tested horizon where the model still explains most of the variance

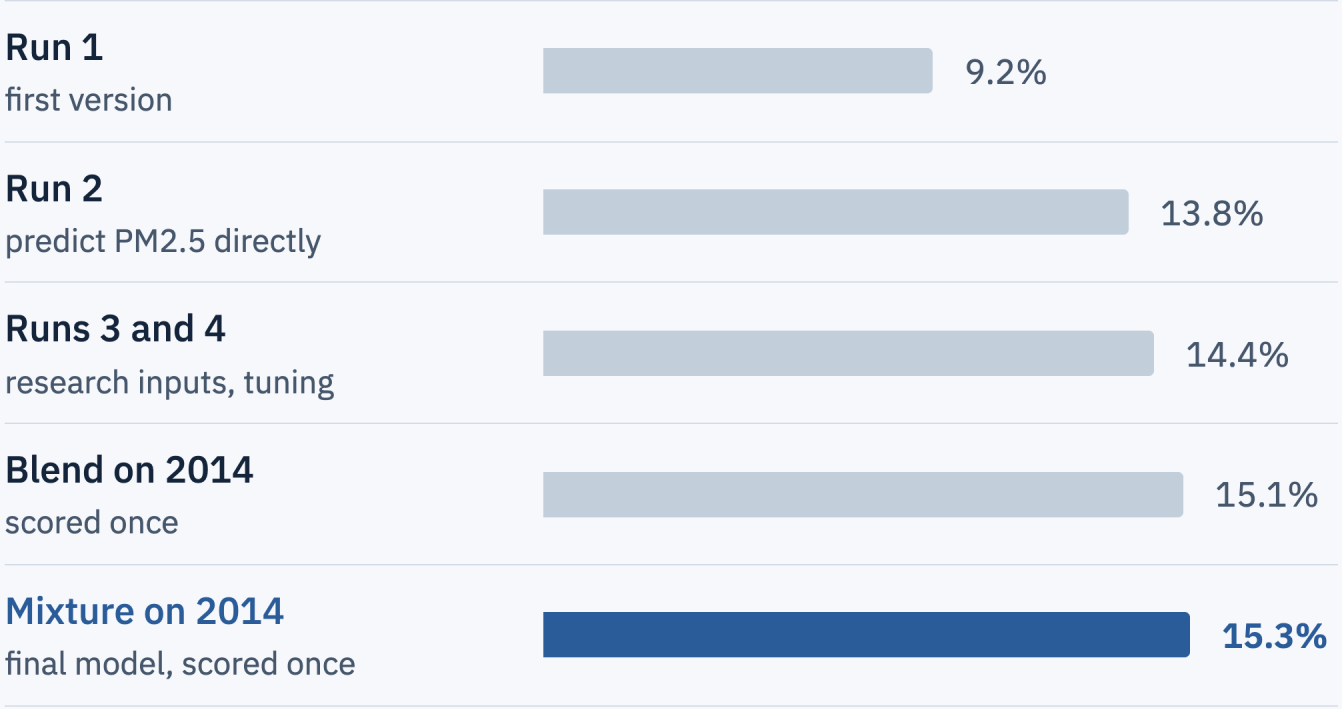
Target form at 6 h, XGBoost, 5 seeds

Target	RMSE	Bias
log1p	59.96	−12.63
log1p + smearing	58.11	+7.10
Raw level	56.81	+1.71
Change from now	58.05	+0.89

Undoing log1p forecasts too low on average; the raw level has the lowest RMSE

Step by step, the model went from 9% to 15% better than the simple guess

% = how much smaller our RMSE is than “no change”, at 6 h



Biggest fix: predict PM2.5 directly

Our first version used a log scale and forecast 12.7 µg/m³ too low on average

A leak caught before the first run

Forecasts from the last 6 hours of 2013 would have been checked against 2014 values; fixed and tested, so no score was affected

Research inputs and tuning

Four inputs from smog research, every setting tuned to cut big misses

On the unseen year 2014, the blend cuts RMSE by 15.1% against persistence

RMSE, µg/m³

51.96

Persistence 61.22

MAE, µg/m³

33.93

Persistence 38.67

R^2, variance explained

0.690

Persistence 0.569

Validation skill was 14.4%, so the gain carried over. On validation, hours above 150 µg/m³ are 21.2% of hours but carry 50.6% of the squared error.

Final model: the soft mix lowers MAE on validation and on 2014; a perfect gate would cut RMSE by 30%

Gate
p = chance PM2.5 at t + 6 exceeds 150

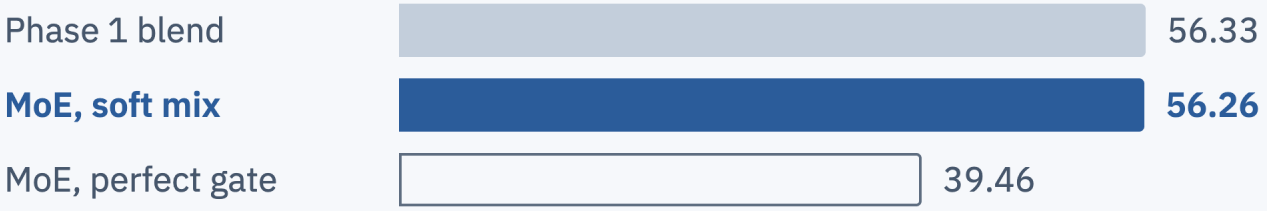
Quiet expert
hours at or below 150

Spike expert
hours above 150

$$(1 - p) \times \text{quiet} + p \times \text{spike}$$

Chosen on 2011 to 2013, then scored once on 2014

RMSE at 6 h, 2011 to 2013, µg/m³



MAE 36.41 to 35.99 on validation, lower in every seed set, and 33.93 to 33.69 on 2014; RMSE level

The gate struggles to see an episode end (AUC 0.734 on hours starting above 150)

Perfect gate uses the true label: a reference, not a forecast

Who built what

Ishan Goyal	EDA on the original data, combined data report
Vedant Madne	Preprocessing pipeline and walkthrough, physics feature code, tests
Pampana Sree Adithya	Preprocessing report and figures, skew check, feature builders, README diagrams
Hrishi Dongre	Baselines, ridge, XGBoost, Phase 1 experiments, results and report
Vetriselvan R	Phase 1 experiments, research for the physics inputs, the MoE model

